

MINI Project

ActivitySelection.java

```
1 import java.util.Scanner;
2
3 public class ActivitySelection {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         System.out.print("Enter number of activities: ");
8         int n = sc.nextInt();
9
10        int[] start = new int[n];
11        int[] finish = new int[n];
12
13        // Input start and finish times
14        for (int i = 0; i < n; i++) {
15            System.out.print("Enter start and finish of activity " + (i + 1) + ": ");
16            start[i] = sc.nextInt();
17            finish[i] = sc.nextInt();
18        }
19
20        // Sort activities by finish time using simple bubble sort
21        for (int i = 0; i < n - 1; i++) {
22            for (int j = i + 1; j < n; j++) {
23                if (finish[i] > finish[j]) {
24                    // Swap finish times
25                    int temp = finish[i];
26                    finish[i] = finish[j];
27                    finish[j] = temp;
28
29                    // Swap corresponding start times
30                    temp = start[i];
31                    start[i] = start[j];
32                    start[j] = temp;
33                }
34            }
35        }
36
37        // Activity Selection
38        System.out.println("\nSelected activities:");
39        int lastFinish = 0;
40        for (int i = 0; i < n; i++) {
41            if (start[i] >= lastFinish) {
42                System.out.println("Activity " + (i + 1) + ": Start=" + start[i] + ", Finish=" + finish[i]);
43                lastFinish = finish[i];
44            }
45        }
46
47        sc.close();
48    }
49 }
```

Output :-

```
Enter number of activities: 5
Enter start and finish of activity 1: 1 3
Enter start and finish of activity 2: 2 4
Enter start and finish of activity 3: 3 5
Enter start and finish of activity 4: 5 9
Enter start and finish of activity 5: 7 6
```

```
Selected activities:
Activity 1: Start=1, Finish=3
Activity 3: Start=3, Finish=5
Activity 4: Start=7, Finish=6
```

```
=== Code Execution Successful ===
```